AUTOMATED INFANT CRY RECOGNITION USING DEEP LEARNING

Automated Infant Cry Recognition System

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SV Signature

Automated Infant Cry Recognition

PROBLEMS

- Difficulty for new parents to understand different types of infant cries
- Noisy environments make cry classification inaccurate
- Lack of accessible and automated tools to analyze and interpret infant needs efficiently.

EXISTING ALTERNATIVES

Manual listening by caregivers and healthcare professionals, subjective and error-prone interpretation of cries

SOLUTION

Instant, trustworthy, and private baby care support powered by AI, tailored to every cry via mobile application

KEY METRIC

1. Cry classification accuracy (target: ≥ 84%)
2. Response time of analysis
3. User adoption and satisfaction rate
4. Number of successful predictions in real-world scenarios

UNFAIR ADVANTAGE

1. Hybrid model combining CNN and LSTM tailored for audio classification in pediatric domain
2. Mobile-based with simple UI for universal caregiver accessibility
3. Uses a diverse dataset from GitHub and Kaggle for better generalization

CHANNELS

- Parenting and health blogs
- Pediatric clinics and hospitals
- App Store & Google Play
- University and tech showcases (e.g., FYP exhibitions)

VALUE PROPOSITIONS

accurate classification of infant cries into categories like hunger, pain, or discomfort using a mobile app powered by deep learning

HIGH-LEVEL CONCEPT

It's like "Google Translate" for babies, decoding cries into needs using AI

CUSTOMER SEGMENTS

1. Parents (new and experienced)
2. Pediatricians and nurses
3. Infant care professionals (e.g., daycare centers)

EARLY ADOPTERS

1. New parents with infants aged 0–2 years
2. Pediatric clinics and NICUs
3. Mobile health app users

COST STRUCTURE

FIXED COSTS:

- UI/UX and mobile app design
- Server/API hosting

VARIABLE COSTS:

- Cloud storage and app maintenance
- Feature updates and support services

REVENUE STREAM

1. Freemium model, basic cry recognition free, premium analytics with subscription
2. Paid partnerships with pediatric clinics or parenting platforms
3. Donations or crowdfunding from parent communities